

Chapter 2. Classification of Mechanical Ventilators and Modes of Ventilation

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Classification of Mechanical Ventilators and Modes of Ventilation: Introduction

A good ventilator classification scheme describes how ventilators work in general terms, but with enough detail so that one particular model can be distinguished from others. It facilitates description by focusing on key attributes in a logical and consistent manner. A clear description allows us to quickly assess new facts in relation to our previous knowledge. Learning the operation of a new ventilator or describing it to others then becomes much easier. Understanding how the ventilator operates, we can then anticipate appropriate ventilator management strategies for particular clinical situations. The classification system described in this chapter is based on previously published work.¹⁻⁷

A ventilator is simply a machine, a system of related elements designed to alter, transmit, and direct energy in a predetermined manner to perform useful work. We put energy into the ventilator in the form of electricity (energy = volts × amps × time) or compressed gas (energy = pressure × volume). That energy is transmitted or transformed (by the ventilator's drive mechanism) in a predetermined manner (by the control circuit) to augment or replace the patient's muscles in performing the work of breathing. Thus to understand mechanical ventilators in general, we must first understand their basic functions: (a) power input, (b) power transmission or conversion, (c) control scheme, and (d) output. This simple format can be expanded to add as much detail as desired ([Table 2-1](#)).

Table 2-1: Outline of Ventilator Classification System

I. Input	IV. Output
A. Pneumatic	A. Pressure waveforms
B. Electric	1. Rectangular
1. AC	2. Exponential
2. DC (battery)	3. Sinusoidal
II. Power conversion and transmission	4. Oscillating
A. External compressor	B. Volume waveforms
B. Internal compressor	1. Ascending ramp
C. Output control valves	2. Sinusoidal
III. Control scheme	C. Flow waveforms
A. Control circuit	1. Rectangular
1. Mechanical	2. Ascending ramp
2. Pneumatic	3. Descending ramp
3. Fluidic	4. Sinusoidal
4. Electric	V. Alarms
5. Electronic	A. Input power alarms
B. Control variables	1. Loss of electric power
1. Pressure	2. Loss of pneumatic power
2. Volume	B. Control circuit alarms
3. Time	1. General systems failure
C. Phase variables	2. Incompatible ventilator settings
1. Trigger	3. Warnings (e.g., inverse inspiratory-to-expiratory timing ratio)
2. Target	C. Output alarms (high/low conditions)
3. Cycle	1. Pressure
4. Baseline	2. Volume
D. Modes of ventilation	3. Flow
1. Control variable	4. Time
2. Breath sequence	a. Frequency
3. Targeting schemes	b. Inspiratory time
	c. Expiratory time
	5. Inspired gas
	a. Temperature
	b. FI_{O_2}

A discussion of input power sources and power conversion and transmission is beyond the scope of this chapter; these topics have been treated elsewhere.^{7,8} The chapter does, however, explore in detail control schemes and ventilator modes because these directly affect patient management.

Control System

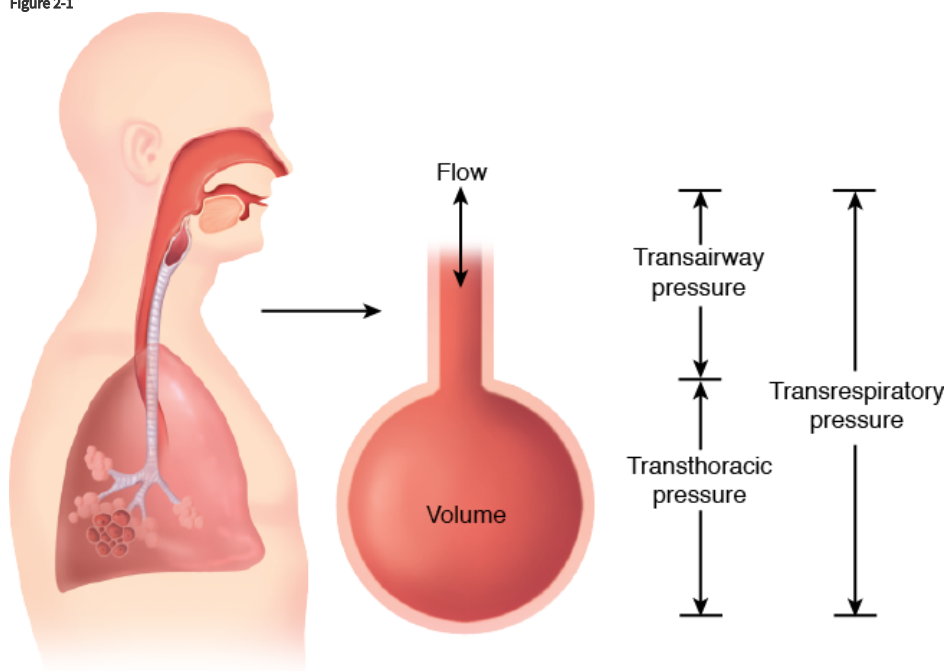
Models of Patient–Ventilator Interaction

To understand how a machine can be controlled to replace or supplement the natural function of breathing, we need to first understand something about the mechanics of breathing itself. The study of mechanics deals with forces, displacements, and the rate of change of displacement. In physiology, force is measured as pressure (pressure = force/area), displacement as volume (volume = area × displacement), and the relevant rate of change as flow [average flow = $\Delta\text{volume}/\Delta\text{time}$; instantaneous flow ($\dot{V}$) = dv/dt , the derivative of volume with respect to time]. Specifically, we are interested in the *pressure* necessary to cause a *flow* of gas to enter the airway and increase the *volume* of the lungs.

The study of respiratory mechanics is essentially the search for simple but useful models of respiratory system mechanical behavior. Figure 2-1 illustrates the process by which the respiratory system is represented first by a graphical model, and then by a mathematical model based on the graphical model. Pressure, volume, and flow are measurable variables in the mathematical model that change with time over the course of one inspiration and expiration. The relation among them is described by the *equation of motion for the respiratory system*.⁹ The derivation of this equation stems from a force-balance equation that is an expression of Newton's third law of motion (for every action, there is an equal and opposite reaction):

$$P_{TR} = P_E + P_R \quad (1)$$

Figure 2-1



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The respiratory system is often modeled as a single flow resistance (representing the endotracheal tube and the airways) connected to an elastic chamber (representing the lungs and chest wall). Flow through the airways is generated by transairway pressure (pressure at the airway opening minus pressure in the lungs). Expansion of the elastic chamber is generated by transthoracic pressure (pressure in the lungs minus pressure on the body surface). Transrespiratory pressure (pressure at the airway opening minus pressure on the body surface) is the sum of these two pressures and is the total pressure required to generate inspiration. The “airway-pressure” gauge on a positive-pressure ventilator displays transrespiratory pressure.

where P_{TR} is the transrespiratory pressure (i.e., pressure at the airway opening minus pressure at the body surface), P_E is the pressure caused by elastic recoil (elastic load), and P_R is the pressure caused by flow resistance (resistive load).

Transrespiratory pressure can have two components, one generated by the ventilator (P_{vent}) and one generated by the respiratory muscles (P_{mus}).

Elastic recoil pressure is the product of elastance ($E = \Delta\text{pressure}/\Delta\text{volume}$) and volume. Resistive pressure is the product of resistance ($R = \Delta\text{pressure}/\Delta\text{flow}$) and flow. Thus, Eq. (1) can be expanded to yield the following equation for inspiration:

$$P_{vent} + P_{mus} = EV + R\dot{V} \quad (2)$$

The combined ventilator and muscle pressure causes volume and flow to be delivered to the patient. (Of course, muscle pressure may subtract rather than add to ventilator pressure in the case of patient-ventilator dyssynchrony, in which case both volume and flow delivery are reduced.) Pressure, volume, and flow are functions of time and are called *variables*. They are all measured relative to their values at end-expiration. Elastance and resistance are assumed to remain constant and are called *parameters*.

For passive expiration, both ventilator and muscle pressure are absent, so Eq. (2) becomes

$$-R\dot{V} = EV \quad (3)$$

The negative sign on the left side of the equation indicates flow in the expiratory direction. This equation also shows that passive expiratory flow is generated by the energy stored in the elastic compartment (i.e., lungs and chest wall) during inspiration.

Equation (2) shows that if the patient’s respiratory muscles are not functioning, muscle pressure is zero, and the ventilator must generate all the pressure for inspiration. On the other hand, a ventilator is not needed for normal spontaneous breathing (i.e., vent pressure = 0). Between those two extremes, an infinite number of combinations of muscle pressure (i.e., patient effort) and ventilator pressure are possible under the general heading of “partial ventilator support.” The equation of motion also gives the basis for defining an *assisted* breath as one for which ventilator pressure rises above baseline during inspiration or falls below baseline during expiration.

Control Variables

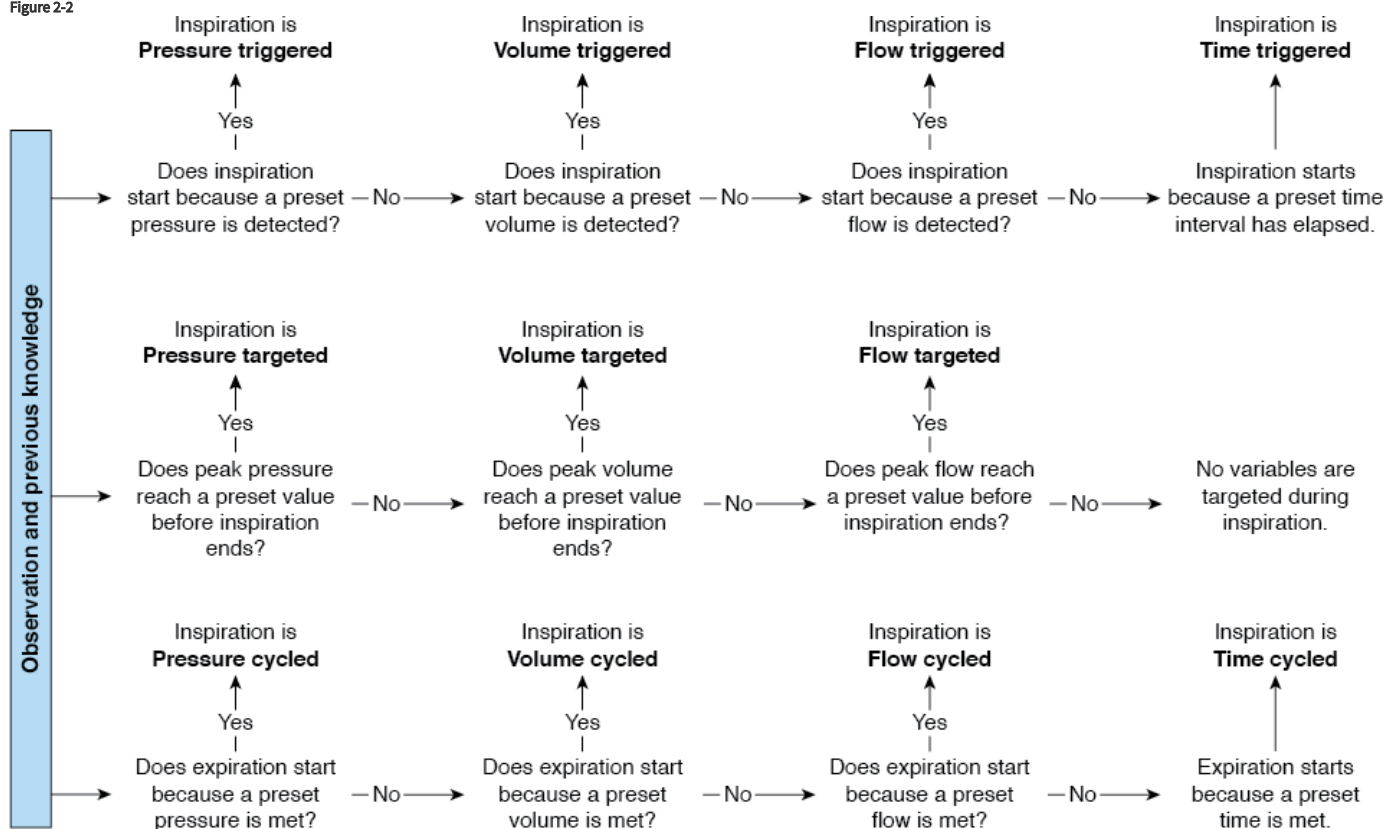
In the equation of motion, the mathematical form of any of the three variables (i.e., pressure, volume, or flow as functions of time) can be predetermined, making it the independent variable and making the other two the dependent variables. We now have a theoretical basis for classifying ventilators as pressure, volume, or flow controllers. Thus, during pressure-controlled ventilation, pressure is the independent variable and may take the form of, say, a step function (i.e., a rectangular pressure waveform). The shapes of the volume and flow waveforms for a passive respiratory system ($P_{mus} = 0$) then depends on the shape of the pressure waveform as well as the parameters of resistance and compliance. On the other hand, during volume-controlled ventilation, we can specify the shape of the volume waveform making flow-dependent and pressure-dependent variables. The same reasoning applies to a flow controller. Notable exceptions are interpulmonary percussive ventilation, and high-frequency oscillatory ventilation, both of which control only the duration of flow pulses; the resulting airway pressure pulses along with actual inspiratory flows and volumes depend on the instantaneous values of respiratory system impedance. Because neither pressure, volume, nor flow in the equation of motion are predetermined, we would classify this type of device as a “time controller.”

It follows from the preceding discussion that any conceivable ventilator can control only one variable at a time: pressure, volume, or flow. Because volume and flow are inverse functions of one another, we can simplify our discussion and consider only pressure and volume as *control variables*. I discuss later in “Modes of Ventilation” exactly how ventilator control systems work. We will see that it is possible for a ventilator to switch quickly from one control variable to another, not only from breath to breath, but even during a single inspiration.

Phase Variables

Because breathing is a periodic event, the ventilator must be able to control a number of variables during the respiratory cycle (i.e., the time from the beginning of one breath to the beginning of the next). Mushin et al¹⁰ proposed that this time span be divided into four phases: the change from expiration to inspiration, inspiration, the change from inspiration to expiration, and expiration. This convention is useful for examining how a ventilator starts, sustains, and stops an inspiration and what it does between inspirations. A particular variable is measured and used to start, sustain, and end each phase. In this context, pressure, volume, flow, and time are referred to as *phase variables*.¹¹ Figure 2-2 shows the criteria for determining phase variables.

Figure 2-2



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Criteria for determining the phase variables during a ventilator-assisted breath.

Trigger Variable

All ventilators measure one or more variables associated with the equation of motion (i.e., pressure, volume, flow, or time). Inspiration is started when one of these variables reaches a preset value. Thus, the variable of interest is considered an initiating, or *trigger*, variable. Time is a trigger variable when

the ventilator starts a breath according to a set frequency independent of the patient's spontaneous efforts. Pressure is the trigger variable when the ventilator senses a drop in baseline pressure caused by the patient's inspiratory effort and begins a breath independent of the set frequency. Flow or volume are the trigger variables when the ventilator senses the patient's inspiratory effort in the form of either flow or volume into the lungs.

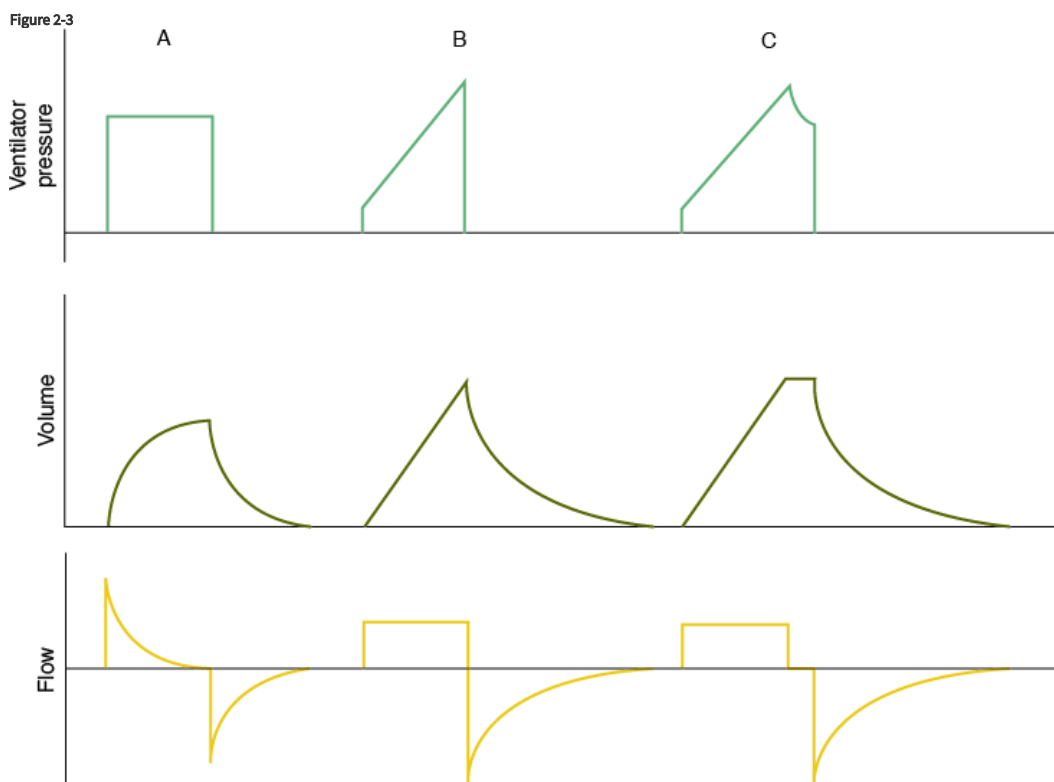
Flow triggering reduces the work the patient must perform to start inspiration.¹² This is so because work is proportional to the volume the patient inspires times the change in baseline pressure necessary to trigger. Pressure triggering requires some pressure change and hence an irreducible amount of work to trigger. With flow or volume triggering, however, baseline pressure need not change, and theoretically, the patient need do no work on the ventilator to trigger.

The patient effort required to trigger inspiration is determined by the ventilator's *sensitivity* setting. Some ventilators indicate sensitivity qualitatively ("min" or "max"). Alternatively, a ventilator may specify a trigger threshold quantitatively (e.g., 5 cm H₂O below baseline). Once the trigger variable signals the start of inspiration, there is always a short delay before flow to the patient starts. This delay is called the *response time* and is secondary to the signal-processing time and the mechanical inertia of the drive mechanisms. It is important for the ventilator to have a short response time to maintain optimal synchrony with patient inspiratory effort.

Target Variable

Here *target* means restricting the magnitude of a variable during inspiration. A *target variable* is one that can reach and maintain a preset level *before* inspiration ends (i.e., it does not end inspiration). Pressure, flow, or volume can be target variables and actually all can be active for a single breath (e.g., using the $P_{\max}$ feature on a Dräger ventilator). Note that time cannot be a target variable because specifying an inspiratory time would cause inspiration to end, violating the preceding definition. Astute readers may notice that in the past I have used the term *limit* where here I have used *target*. This was done to be consistent with the International Standards Organization's use of the term *limit* as applying to alarm situations only.

Clinicians often confuse target variables with cycle variables. To *cycle* means "to end inspiration." A cycle variable always ends inspiration. A target variable does not terminate inspiration; it only sets an upper bound for pressure, volume, or flow (Fig. 2-3).



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This figure illustrates the distinction between the terms *target* and *cycle*. **A.** Inspiration is pressure-targeted and time-cycled. **B.** Flow is targeted, but volume is not, and inspiration is volume-cycled. **C.** Both volume and flow are targeted, and inspiration is time-cycled. (Reproduced, with permission, from Chatburn.⁶)

Cycle Variable

The inspiratory phase always ends when some variable reaches a preset value. The variable that is measured and used to end inspiration is called the *cycle variable*. The cycle variable can be pressure, volume, flow, or time. Manual cycling is also available on some ventilators.

When a ventilator is set to pressure cycle, it delivers flow until a preset pressure is reached, at which time inspiratory flow stops and expiratory flow begins. The most common application of pressure cycling on mechanical ventilators is for alarm settings.

When a ventilator is set to volume cycle, it delivers flow until a preset volume has passed through the control valve. By definition, as soon as the set volume is met, inspiratory flow stops and expiratory flow begins. If expiration does not begin immediately after inspiratory flow stops, then an inspiratory hold has been set, and the ventilator is, by definition, time cycled (see [Fig. 2-3](#)). Note that the volume that passes through the ventilator's output control valve is never exactly equal to the volume delivered to the patient because of the volume compressed in the patient circuit. Some ventilators use a sensor at the Y-connector (such as the Dräger Evita 4 with the neonatal circuit) for more accurate tidal volume measurement. Others measure volume at some point inside the ventilator, and the operator must know whether the ventilator compensates for compressed gas in its tidal volume readout.

When a ventilator is set to flow cycle, it delivers flow until a preset level is met. Flow then stops, and expiration begins. The most frequent application of flow cycling is in the pressure-support mode. In this mode, the control variable is pressure, and the ventilator provides the flow necessary to meet the inspiratory pressure target. In doing so, flow starts out at a relatively high value and decays exponentially (assuming that the patient's respiratory muscles are inactive after triggering). Once flow has decreased to a relatively low value (such as 25% of peak flow, typically preset by the manufacturer), inspiration is cycled off. Manufacturers often set the cycle threshold slightly above zero flow to prevent inspiratory times from getting so long that patient synchrony is degraded. On some ventilators, the flow-cycle threshold may be adjusted by the operator to improve patient synchrony. Increasing the flow-cycle threshold decreases inspiratory time and vice versa.

Time cycling means that expiratory flow starts because a preset inspiratory time interval has elapsed.

Baseline Variable

The baseline variable is the parameter controlled during expiration. Although pressure, volume, or flow could serve as the baseline variable, pressure control is the most practical and is implemented by all modern ventilators. Baseline or expiratory pressure is always measured and set relative to atmospheric pressure. Thus, when we want baseline pressure to equal atmospheric pressure, we set it to zero. When we want baseline pressure to exceed atmospheric pressure, we set a positive value, called *positive end-expiratory pressure* (PEEP).

Modes of Ventilation

The general goals of mechanical ventilation are to promote safety, comfort, and liberation ([Table 2-2](#)).¹ Specific objectives under these goals include ensuring adequate gas exchange, avoiding ventilator induced lung injury, optimizing patient-ventilator synchrony, and minimizing the duration of ventilation. The preset pattern of patient-ventilator interaction designed to achieve these objectives is referred to as a mode of ventilation. Specifically, a mode can be classified according to the outline in [Table 2-3](#).²

Table 2-2: Goals and Objectives of Mechanical Ventilation.

1. Promote safety
a. Optimize ventilation–perfusion of the lung
i. Maximize alveolar ventilation
ii. Minimize shunt
b. Optimize pressure–volume curve
i. Minimize tidal volume
ii. Maximize compliance
2. Promote comfort
a. Optimize patient–ventilator synchrony
i. Maximize trigger–cycle synchrony
ii. Minimize auto-PEEP
iii. Maximize flow synchrony
iv. Coordinate mandatory and spontaneous breaths
b. Optimize work demand versus work delivered
i. Minimize inappropriate shifting of work from ventilator to patient
3. Promote liberation
a. Optimize the weaning experience
i. Minimize adverse events
ii. Minimize duration of ventilation

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Table 2-3: Outline of Mode Classification System

1. Primary control variable
a. Pressure
b. Volume
2. Breath sequence
a. Continuous mandatory ventilation (CMV)
b. Intermittent mandatory ventilation (IMV)
c. Continuous spontaneous ventilation (CSV)
3. Primary targeting scheme
a. Set-point
b. Dual
c. Servo
d. Adaptive
e. Optimal
f. Intelligent
4. Secondary targeting scheme
a. Set-point
b. Servo
c. Adaptive
d. Optimal
e. Intelligent

Control Variable

I have already mentioned that pressure, volume, or flow can be controlled during inspiration. When discussing modes I will refer to inspiration as being pressure-controlled or volume-controlled. Ignoring flow control is justified because when the ventilator controls volume directly (i.e., using a volume-

feedback signal), flow is controlled indirectly, and vice versa (i.e., mathematically, volume is the integral of flow, and flow is the derivative of volume).

There are clinical advantages and disadvantages to volume and pressure control. To keep within the scope of this chapter, we can just say that volume control results in a more stable minute ventilation (and hence more stable blood gases) than pressure control if lung mechanics are unstable. On the other hand, pressure control allows better synchronization with the patient because inspiratory flow is not constrained to a preset value. Although the ventilator must control only one variable at a time during inspiration, it is possible to begin a breath-in pressure control and (if certain criteria are met) switch to volume control or vice versa (referred to as dual targeting, described in “[Targeting Schemes](#)” below).

Breath Sequence

The *breath sequence* is the pattern of mandatory or spontaneous breaths that the mode delivers. A *breath* is a positive airway flow (inspiration) relative to baseline, and it is paired (by size) with a negative airway flow (expiration), both associated with ventilation of the lungs. This definition excludes flow changes caused by hiccups or cardiogenic oscillations. It allows, however, the superimposition of, for example, a spontaneous breath on a mandatory breath or vice versa. The flows are paired by size, not necessarily by timing. In airway pressure-release ventilation, for example, there is a large inspiration (transition from low pressure to high pressure) possibly followed by a few small inspirations and expirations, followed finally by a large expiration (transition from high pressure to low pressure). These comprise several small spontaneous breaths superimposed on one large mandatory breath. During high-frequency oscillatory ventilation, in contrast, small mandatory breaths are superimposed on larger spontaneous breaths.

A *spontaneous breath*, in the context of mechanical ventilation, is a breath for which the patient determines both the timing and the size. The start and end of inspiration may be determined by the patient, independent of any machine settings for inspiratory time and expiratory time. That is, the patient both triggers and cycles the breath. On some ventilators, the patient may make short, small spontaneous efforts during a longer, larger mandatory breath, as in the case of *airway pressure-release ventilation*. It is important to make a distinction between spontaneous breaths and assisted breaths. An *assisted breath* is one for which the ventilator does some work for the patient, as indicated by an increase in airway pressure (i.e., P_{vent}) above baseline during inspiration or below baseline during expiration. For example, in the *pressure-support mode*, each breath is assisted because airway pressures rise to the *pressure-support* setting above PEEP (i.e., $P_{vent} > 0$). Each breath is also spontaneous because the patient both triggers and cycles the breath. The patient may cycle the breath in the *pressure-support* mode by actively exhaling, but even if the patient is passive at end-inspiration, the patient's resistance and compliance determine the cycle point and thus the size of the breath for a given *pressure-support* setting. In contrast, for a patient on continuous positive airway pressure, each breath is spontaneous but unassisted. Breaths are spontaneous because the patient determines the timing and size of the breaths without any interference by the ventilator. Breaths during continuous positive airway pressure are not assisted because airway pressure is controlled by the ventilator to be as constant as possible (i.e., $P_{vent} = 0$). Understanding the difference between assisted and unassisted spontaneous breaths is very important clinically. When making measurements of tidal volume and respiratory rate for calculation of the rapid-shallow breathing index, for example, the breaths must be *spontaneous* and *unassisted*. If they are assisted (e.g., with pressure support), an error of 25% to 50% may be introduced.

A *mandatory breath* is any breath that does not meet the criteria of a spontaneous breath, meaning that the patient has lost control over the timing and/or size. Thus, a mandatory breath is one for which the start or end of inspiration (or both) is determined by the ventilator, independent of the patient; that is, the machine triggers and/or cycles the breath. It is possible to superimpose a short mandatory breath on top of a longer spontaneous breath, as in the case of high-frequency oscillatory ventilation.

Having defined spontaneous and mandatory breaths, there are three possible breath sequences, designated as follows:

Continuous spontaneous ventilation (CSV). All breaths are spontaneous.

Intermittent mandatory ventilation (IMV). Spontaneous breaths are permitted between mandatory breaths. When the mandatory breath is triggered by the patient, it is commonly referred to as *synchronized IMV*. Because the trigger variable can be specified in the description of phase variables, I will use IMV instead of synchronized IMV to designate general breath sequences.

Continuous mandatory ventilation (CMV). Spontaneous breaths are not permitted between mandatory breaths, as the intent is to provide a mandatory breath for every patient inspiratory effort. *CMV* originally meant that every breath was mandatory. The development of the “active exhalation valve,” however, made it possible for the patient to breathe spontaneously during a mandatory pressure-controlled breath on some ventilators. In fact, it was always possible for the patient to breathe spontaneously during pressure-controlled mandatory breaths on infant ventilators. *The key distinction between CMV and IMV is that with CMV, the ventilator attempts to deliver a mandatory breath every time the patient makes an inspiratory effort* (unless a mandatory breath is already in progress). This means that during *CMV*, if the operator decreases the ventilator rate, the level of ventilator support is unaffected as long as the patient continues making inspiratory efforts. With IMV, the rate setting directly affects the number of mandatory breaths and hence the level of ventilator support. Thus, *CMV* is normally viewed as a method of “full ventilator support,” whereas IMV is usually viewed as a method of partial ventilator support. Of course, actual “full ventilatory support” can only be achieved if the patient is making no inspiratory efforts, for example, is paralyzed, but the term is often used loosely to mean supplying as much support as possible for a given patient condition.

Given the two ways to control inspiration (i.e., pressure and volume) and the three breath sequences (i.e., *CMV*, IMV, or CSV), there are five possible breathing patterns; volume control (VC)-*CMV*, VC-IMV, pressure control (PC)-*CMV*, PC-IMV, PC-CSV (see [Table 2-2](#)). VC-CSV is not possible because volume

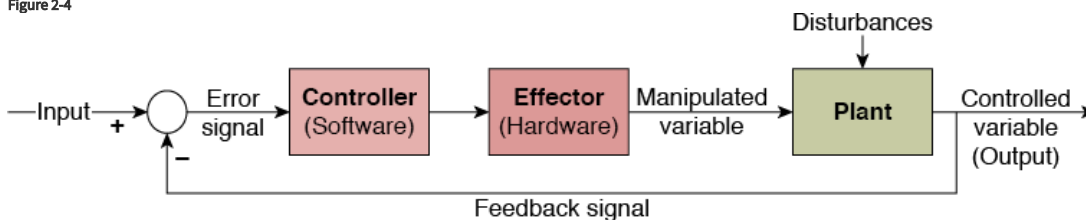
control implies that inspiration ends after a preset tidal volume is delivered, hence violating the patient cycling criterion of a spontaneous breath.

Targeting Schemes

Targeting schemes are feedback control systems used by mechanical ventilators to deliver specific ventilatory patterns.¹ The targeting scheme is a key component of a mode classification system. Before we can describe specific targeting schemes used by ventilators, we must first appreciate the basic concepts of engineering control theory.

The term *closed-loop control* refers to the use of a feedback signal to adjust the output of a system. Ventilators use closed-loop control to maintain consistent pressure and flow waveforms in the face of changing patient/system conditions. This is accomplished by using the output as a feedback signal that is compared to the operator-set input. The difference between the two is used to drive the system toward the desired output. For example, pressure-control modes use airway pressure as the feedback signal to control gas flow from the ventilator. Figure 2-4 is a schematic of a general control system. The input is a reference value (e.g., operator preset inspiratory pressure) that is compared to the actual output value (e.g., instantaneous value of airway pressure). The difference between those two values is the error signal. The error signal is passed to the controller (e.g., the software control algorithm). The controller converts the error signal into a signal that can drive the effector (e.g., the hardware) to cause a change in the manipulated variable (e.g., inspiratory flow). The relationship between the input and the output of the controller is called the transfer function in control theory. Engineers need to understand the transfer function in terms of complex mathematical equations. Clinicians, however, need only understand the general operation of the function in terms of how the mode affects the patient's ventilatory pattern, and we will use that frame of reference in defining targeting schemes. The "plant" in Figure 2-4 refers to the process under control. In our case, the plant is the patient and the delivery circuit connecting the patient to the ventilator. The plant is the source of the "noise" that causes problems with patient-ventilator synchrony. At one extreme, a paralyzed patient and an intact delivery circuit pose little challenge for a modern ventilator to deliver a predetermined ventilatory pattern, and thus synchrony is not an issue. At the opposite extreme is a patient with an intense, erratic respiratory drive and a delivery circuit with leaks (e.g., around an uncuffed endotracheal tube) making patient-ventilator synchrony virtually impossible. The challenge for both clinicians and engineers is to develop technology and procedures for dealing with this wide range of circumstances.

Figure 2-4



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Generalized control circuit (see text for explanation). The "plant" in a control circuit for mechanical ventilation is the patient. (Reproduced with permission from Chatburn RL, Mireles-Cabodevila E, Closed loop control of mechanical ventilation. *Respir Care*. 2011;56(1):85–98.)

The plant alters the manipulated variable to generate the feedback signal of interest as the control (output) variable. Continuing with the example above, the manipulated variable is flow, but the feedback control variable is pressure (i.e., ventilator flow times plant impedance equals airway pressure), as in pressure-control modes.

Closed-loop control can also refer to the use of feedback signals to control the overall pattern of ventilation, beyond a single breath, such as the use of end-tidal carbon dioxide tension as a feedback signal to control minute ventilation.

The process of "setting" or adjusting a ventilation mode can be thought of as presetting various target values, such as tidal volume, inspiratory flow, inspiratory pressure, inspiratory time, frequency, PEEP, oxygen concentration, and end-tidal carbon dioxide concentration. The term *target* is used for two reasons. First, just like in archery, a target is aimed at but not necessarily hit, depending on the precision of the control system. An example is setting a target value for tidal volume and allowing the ventilator to adjust the inspiratory pressure over several breaths to finally deliver the desired value. In this case, we could more accurately talk about delivering an average target tidal volume over time.

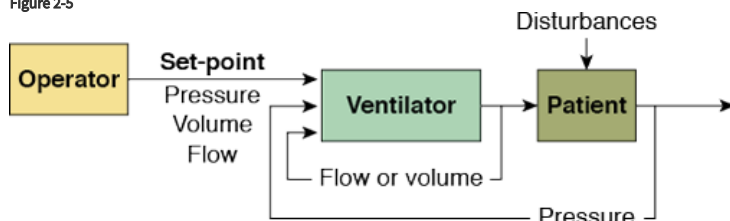
The second reason for using *target* is because the term *control* is overused and we need it to preserve some fundamental conventions regarding modes such as *volume control* versus *pressure control*. From this use of the term *target*, we can logically refer to the control system transfer function (relationship between the input and the output of the controller) as a targeting scheme. The history of these schemes clearly shows an evolutionary trend toward increasing levels of automation. In fact, we can identify three groups of targeting schemes based on increasing levels of autonomy: manual, servo, and automatic. Manual targeting schemes require the operator to adjust all the target values. Servo targeting schemes are unique in that there are no static target values; rather, the operator sets the parameters of a mathematical model that drives the ventilator's output to follow a dynamic signal (like power steering on an automobile). Automatic targeting schemes enable the ventilator to set some or all of the ventilatory targets, using either mathematical models of physiologic processes or artificial-intelligence algorithms.

The basic concept of closed-loop control has evolved into at least six different ventilator targeting schemes (set-point, dual, servo, adaptive, optimal, and intelligent). These targeting schemes are the foundation that makes possible several dozen apparently different modes of ventilation. Once we understand how these control types work, many of the apparent differences are seen to be similarities. We then avoid a lot of the confusion surrounding ventilator marketing hype and begin to appreciate the true clinical capabilities of different ventilators.

Set-Point

In set-point targeting, the operator sets specific target values and the ventilator attempts to deliver them (Fig. 2-5). The simplest examples for volume-control modes are tidal volume and inspiratory flow. For pressure-control modes, the operator may set inspiratory pressure and inspiratory time or cycle threshold.

Figure 2-5



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Set-point targeting. (Reproduced, with permission, from Chatburn RL. Computer control of mechanical ventilation. *Respir Care*. 2004;49:507–515.)

Dual

As it relates to mechanical ventilation, volume control means that inspired volume, as a function of time, is predetermined by the operator before the breath begins. In contrast, pressure control means that inspiratory pressure as a function of time is predetermined. “Predetermined” in this sense means that either pressure or volume is constrained to a specific mathematical form. In the simple case where either pressure or flow are preset constant values (e.g., set-point targeting, as explained above), we can say that they are the independent variables in the equation of motion. The equation of motion for the respiratory system is a general mathematical model of patient–ventilator interaction:

$$P(t) = EV(t) + R\dot{V}(t) \quad (4)$$

where $P(t)$ is inspiratory pressure as a function of time (t), E is respiratory-system elastance, $V(t)$ is volume as a function of time, R is respiratory-system resistance, and $\dot{V}$ is flow as a function of time. Thus, for example, if pressure is the independent variable, then both volume and flow are dependent variables, indicating pressure control. If volume is the independent variable, then pressure is the dependent variable, indicating volume control. Because volume is the integral of flow, if $\dot{V}$ is predetermined, then so is $V(t)$. Therefore, for simplicity, we include the case of flow being the independent variable as a form of volume control.

Only one variable (i.e., pressure or volume) can be independent at any moment, but a ventilator controller can switch between the two during a single inspiration. When this happens, the targeting scheme is called dual set-point control or dual targeting. There are two basic ways that ventilators have implemented dual targeting. One way is to start inspiration in volume control and then switch to pressure control if one or more preset thresholds are met (e.g., a desired peak airway pressure target). An example of such a threshold is the operator-set $P_{\max}$ in volume control on the Dräger Evita XL ventilator. The other form of dual targeting is to start inspiration in pressure control and then switch to volume control (e.g., if a preset tidal volume has not been met when flow decays to a preset value). This was originally described as “volume-assured pressure-support ventilation,”¹³ but is currently only available as a mode called “Volume Control Assist Control with Machine Volume” in the CareFusion Avea ventilator.

Dual targeting is an attempt to improve the synchrony between patient and ventilator. This can be seen in the equation of motion if a term representing the patient inspiratory force (muscle pressure or P_{mus}) is added:

$$P(t) = EV(t) + R\dot{V}(t) - P_{\text{mus}}(t) \quad (5)$$

With set-point targeting in volume control modes, volume and flow are preset. Therefore, if the patient makes an inspiratory effort (i.e., $P_{\text{mus}}(t) > 0$), then the equation dictates that transrespiratory-system pressure, $P(t)$, must fall. Because work is the result of both pressure and volume delivery (i.e., $\text{work} = \int P dV$), if pressure decreases, the work the ventilator does on the patient decreases and hence we have asynchrony of work demand on the part of the patient versus work output on the part of the ventilator.

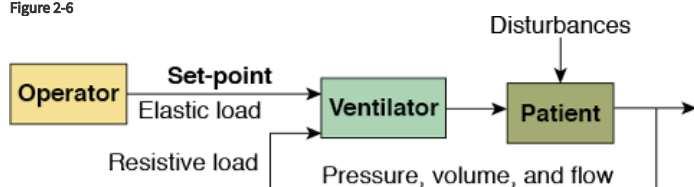
With set-point pressure control, transrespiratory pressure is preset. Consequently, if the patient makes an inspiratory effort, both volume and flow increase. With constant pressure and increased volume, work per liter for the breath stays constant. Although this gives better work synchrony than does

volume control, it is not ideal. Nevertheless, merging of volume and pressure control using a dual targeting scheme provides the safety of a guaranteed minimum tidal volume with the patient comfort of flow synchrony provided by pressure control.

Servo

The term *servo* was coined by Joseph Farcot in 1873 to describe steam-powered steering systems. Later, hydraulic “servos” were used to position anti-aircraft guns on warships. Servo control specifically refers to a control system that converts a small mechanical motion into one requiring much greater power, using a feedback mechanism. As such, it offers a substantial advantage in terms of creating ventilation modes capable of a high degree of synchrony with patient breathing efforts. That is, ventilator work output can be made to match patient work demand with a high degree of fidelity. We apply the name *servo control* to targeting schemes in which the ventilator’s output automatically follows a varying input. This includes proportional-assist ventilation (PAV; Fig. 2-6),¹⁴ automatic tube compensation (ATC),¹⁵ and neurally adjusted ventilatory assist (NAVA),¹⁶ in which the airway pressure signal not only follows but amplifies signals that are surrogates for patient effort (i.e., volume, flow, and diaphragmatic electrical signals). Note that the term *servo control* has been loosely used since it was coined to refer to any type of general feedback control mechanism, but I am using it in a very specific manner, as it applies to ventilator targeting schemes.

Figure 2-6



$$P_{mus} = Load_{normal} + Load_{disease}$$

$$P_{vent} = K_1 \times V + K_2 \times \dot{V}$$

$$P_{mus} + P_{vent} = Load_{normal} + Load_{disease}$$

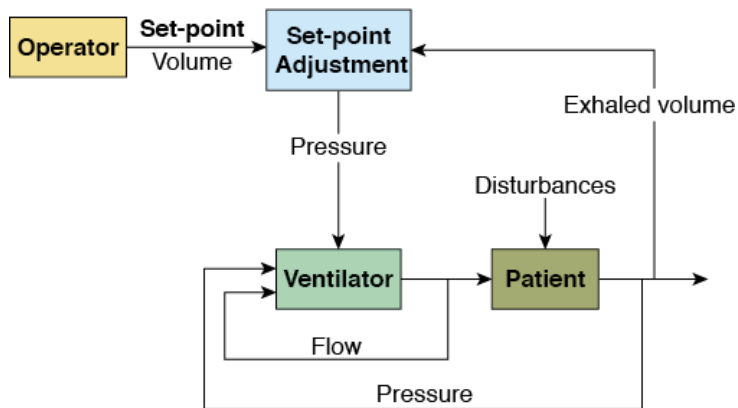
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Servo targeting is the basis for the proportional-assist mode. In this mode, the operator sets targets for elastic and resistive unloading. The ventilator then delivers airway pressure in proportion to the patient’s own inspiratory volume and flow. When the patient’s muscles have to contend with an abnormal load secondary to disease, proportional assist allows the operator to set amplification factors (K_1 and K_2) on the feedback volume and flow signals. By amplifying volume and flow, the ventilator generates a pressure that supports the abnormal load, freeing the respiratory muscles to support only the normal load caused by the natural elastance and resistance of the respiratory system. (Reproduced, with permission, from Chatburn RL. Computer control of mechanical ventilation. *Respir Care*. 2004;49:507–515.)

Adaptive

An adaptive targeting scheme involves modifying the function of the controller to *cope* with the fact that the system parameters being controlled are time varying. As it applies to mechanical ventilation, adaptive targeting schemes allow the ventilator to set some (or conceivably all) of the targets in response to varying patient conditions. Modern intensive care unit ventilators may use adaptive flow targeting as a more accurate way to deliver volume control modes than set-point targeting. For example, the Covidien PB 840 ventilator automatically adjusts inspiratory flow between breaths to compensate for volume compression in the patient circuit and thus achieving an average target tidal volume equal to the operator-set value.¹⁷ Aside from this application of adaptive targeting, there are four distinct approaches to basic adaptive targeting, which are represented by the mode names *pressure-regulated volume control* (inspiratory pressure automatically adjusted to achieve an average tidal volume target, Fig. 2-7), *mandatory rate ventilation* (inspiratory pressure automatically adjusted to maintain a target spontaneous breath frequency), *adaptive flow/adaptive I-time* (inspiratory time and flow automatically adjusted to maintain a constant inspiratory time-to-expiratory time ratio of 1:2), and *mandatory minute ventilation* (automatic adjustment of mandatory breath frequency to maintain a target minute ventilation).

Figure 2-7



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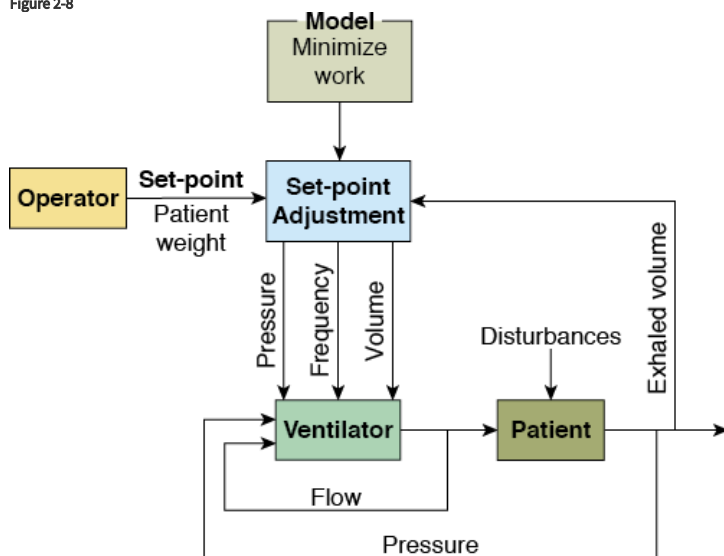
Adaptive targeting. Notice that the operator has stepped back from direct control of the within-breath parameters of pressure and flow. Examples of adaptive targeting are pressure-regulated volume control (PRVC) on the Siemens ventilator and autoflow on the Dräger Evita 4 ventilator. (Reproduced, with permission, from Chatburn RL. Computer control of mechanical ventilation. *Respir Care*. 2004;49:507–515.)

Optimal

Optimal targeting is an advanced form of adaptive targeting.¹⁸ Optimal targeting in this context means that the ventilator controller automatically adjusts the targets of the ventilatory pattern to either minimize or maximize some overall performance characteristic (Fig. 2-8). Adaptive-support ventilation (ASV) on the Hamilton ventilators is the only commercially available mode to date that uses optimal targeting. This targeting scheme was first described by Tehrani in 1991²⁰ and was designed to minimize the work rate of breathing, mimic natural breathing, stimulate spontaneous breathing, and reduce weaning time.²⁰ The operator inputs the patient's weight. From that, the ventilator estimates the required minute alveolar ventilation, assuming a normal dead space fraction. Next, an optimum frequency is calculated based on work by Otis et al²¹ that predicts a frequency resulting in the least mechanical work rate.²⁰

$$f = \frac{-1 + \sqrt{1 + 4\pi^2 RC_E \left(\frac{MV - fV_D}{V_D} \right)}}{2\pi^2 RC_E} \quad (6)$$

Figure 2-8



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Optimal targeting. A static mathematical model is used to optimize some performance parameter, such as work of breathing. The only commercially available form of optimal targeting is the adaptive-support ventilation (ASV) mode on the Hamilton Galileo ventilator. (Reproduced, with permission, from Chatburn RL. Computer control of mechanical ventilation. *Respir Care*. 2004;49:507–515.)

where MV is predicted minute ventilation (L/min) based on patient weight and the setting for percent of predicted MV to support, V_D is predicted dead space (L) based on patient weight, RC_E is the expiratory time constant calculated as the slope of the expiratory flow volume curve and f is the computed optimal frequency (breaths/min). The target tidal volume is calculated as MV/f . The ASV controller uses the Otis equation to set the tidal volume (Fig. 2-8). As with simple adaptive pressure targeting, the inspiratory pressure within a breath is controlled to achieve a constant value and between breaths the inspiratory pressure is adjusted to achieve a target tidal volume. Unlike simple adaptive pressure targeting, however, the target is not set by the operator; instead, it is estimated by the ventilator in response to changes in respiratory-system mechanics and patient effort. Individual pressure-targeted breaths may be mandatory (time triggered and time cycled) or spontaneous (flow triggered and flow cycle).

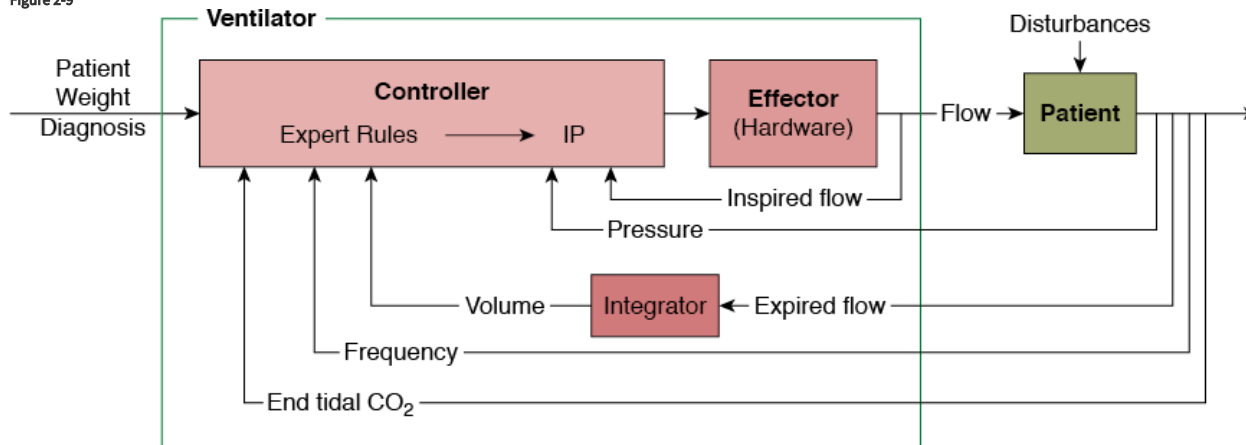
ASV adds some expert rules that put safety limits on frequency and tidal volume delivery and reduce the risk of auto-PEEP. In that sense, this mode may be considered an intelligent targeting scheme, or more appropriately, a hybrid system (i.e., using a mathematical model and artificial intelligence).

Intelligent

Intelligent targeting systems are another form of adaptive targeting schemes that use artificial-intelligence techniques.²² The most convincing proof of the concept was presented by East et al,²³ who used a rule-based expert system for ventilator management in a large, multicenter, prospective, randomized trial. Although survival and length of stay were not different between human and computer management, computer control resulted in a significant reduction in multiorgan dysfunction and a lower incidence and severity of lung overdistension injury. The most important finding, however, was that *expert knowledge can be encoded and shared successfully with institutions that had no input into the model*. Note that the expert system did not control the ventilator directly, but rather made suggestions for the human operator. In theory, of course, the operator could be eliminated.

There is only one ventilator mode commercially available to date in the United States with a targeting scheme that relies entirely on a rule-based expert system (Fig. 2-9). That mode is SmartCare/PS on the Dräger Evita XL ventilator. This mode is a specialized form of pressure support that is designed for true (ventilator led) automatic weaning of patients. The SmartCare/PS controller uses predefined acceptable ranges for spontaneous breathing frequency, tidal volume, and end-tidal carbon dioxide tension to automatically adjust the inspiratory pressure to maintain the patient in a “respiratory zone of comfort.”²³

Figure 2-9



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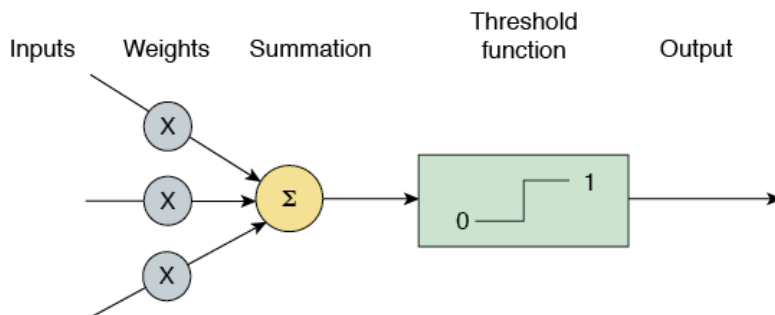
An intelligent targeting system for automatically adjusting pressure support levels (e.g., SmartCare/PS). IP, inspiratory pressure. (Reproduced, with permission, from Chatburn RL, Mireles-Cabodevila E. Closed loop control of mechanical ventilation. *Respir Care*. 2011;56(1):85–98.

The SmartCare/PS system divides the control process into three steps. The first step is to stabilize the patient within the “zone of respiratory comfort” defined as combinations of tidal volume, respiratory frequency, and end tidal CO_2 values defined as acceptable by the artificial-intelligence program. There are different combinations depending on whether the patient has chronic obstructive pulmonary disease or a neuromuscular disorder. The second step is to progressively decrease the inspiratory pressure while making sure the patient remains in the “zone.” The third step tests readiness for extubation by maintaining the patient at the lowest level of inspiratory pressure. The lowest level depends on the type of artificial airway (endotracheal tube vs. tracheostomy tube), the type of humidifier (heat and moisture exchanger vs. a heated humidifier), and the use of automatic tube compensation. Once the lowest level of inspiratory pressure is reached, a 1-hour observation period is started (i.e., a spontaneous breathing trial) during which the patient’s breathing frequency, tidal volume, and end-tidal CO_2 are monitored. Upon successful completion of this step, a message on the screen suggests that the clinician “consider separation” of the patient from the ventilator. This method for automatic weaning reduces the duration of mechanical ventilation and intensive care unit length of stay in a multicenter randomized controlled trial.^{24,25} The advantage of artificial intelligence, however, may be less noticeable in environments where natural intelligence is plentiful. Rose et al recently concluded that “Substantial reductions in weaning duration

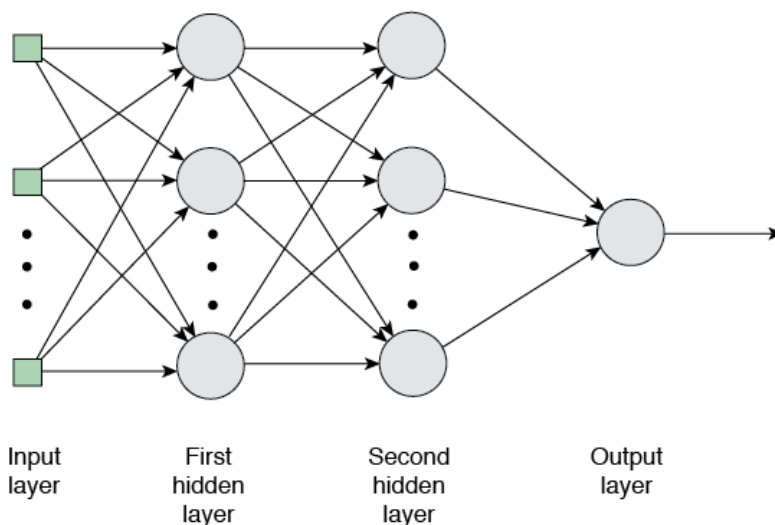
previously demonstrated were not confirmed when the SmartCare/PS system was compared to weaning managed by experienced critical care specialty nurses, using a 1:1 nurse-to-patient ratio. The effect of SmartCare/PS may be influenced by the local clinical organizational context.”²⁶

The ultimate in ventilator targeting system to date is the artificial neural network (Fig. 2-10).²⁷ Again, this experimental system does not control the ventilator directly but acts as a decision-support system. What is most interesting is that *the neural network is capable of learning*, which offers significant advantages over static mathematical models and even expert rule-based systems.

Figure 2-10
Single neuron



Neural network



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Neural network structure. A single neuron accepts inputs of any value and weights them to indicate the strength of the synapse. The weighted signals are summed to produce an overall unit activation. If this activation exceeds a certain threshold, the unit produces an output response. A network is made up of layers of individual neurons. (Reproduced, with permission, from Chatburn RL. Computer control of mechanical ventilation. *Respir Care*. 2004;49:507–515.)

Neural nets are essentially data-modeling tools used to capture and represent complex input–output relationships. A neural net learns by experience the same way a human brain does, by storing knowledge in the strengths of internode connections. As data-modeling tools, they have been used in many business and medical applications for both diagnosis and forecasting.²⁸ A neural network, like an animal brain, is made up of individual neurons. Signals (action potentials) appear at the unit’s inputs (synapses). The effect of each signal may be approximated by multiplying the signal by some number or weight to indicate the strength of the signal. The weighted signals then are summed to produce an overall unit activation. If this activation exceeds a certain threshold, the unit produces an output response. Large numbers of neurons can be linked together in layers (see Fig. 2-10). The nodes in the diagram represent the summation and transfer processes. Note that each node contains information from all neurons. As the network learns, the weights change, and thus the values at the nodes change, affecting the final output.

In summary, ventilator control schemes display a definite hierarchy of evolutionary complexity. At the most basic level, control is focused on what happens within a breath. We can call this *manual control*, and there is a very direct need for operator input of static set-points. The next level up is what we can call *automatic control*. Here, set-points are dynamic in that they may be adjusted automatically over time by the ventilator according to some model of desired performance. The operator is somewhat removed in that inputs are entered at the level of the model and take effect over several breaths instead of at the level of individual breath control. Finally, the highest level so far is what might be considered *intelligent control*. Here, the

operator can be eliminated altogether. Not only dynamic set-points but also dynamic models of desired performance are permitted. There is the possibility of the system learning from experience so that the control actually spans between patients instead of just between breaths.

Mode Classification

When Mushin et al wrote the classic book on automatic ventilation of the lungs,¹⁰ the emphasis was on classifying ventilators and there were very few modes on each device. These devices have undergone a tremendous technological evolution during the intervening years. As a result, there are now more than 170 names of modes on ventilators in the United States alone, with as many as two dozen available on a single device. The proliferation of names makes education of end users very difficult, potentially compromising the quality of patient care. In addition, although there may be more than 170 mode names, these are not uniquely different modes. Consequently, the emphasis today in describing ventilators must be on classifying modes, shifting awareness from names to tags. Much has been written on the subject,^{2,5, 29–31} and this section gives a brief overview of the development and application of a ventilator mode taxonomy.

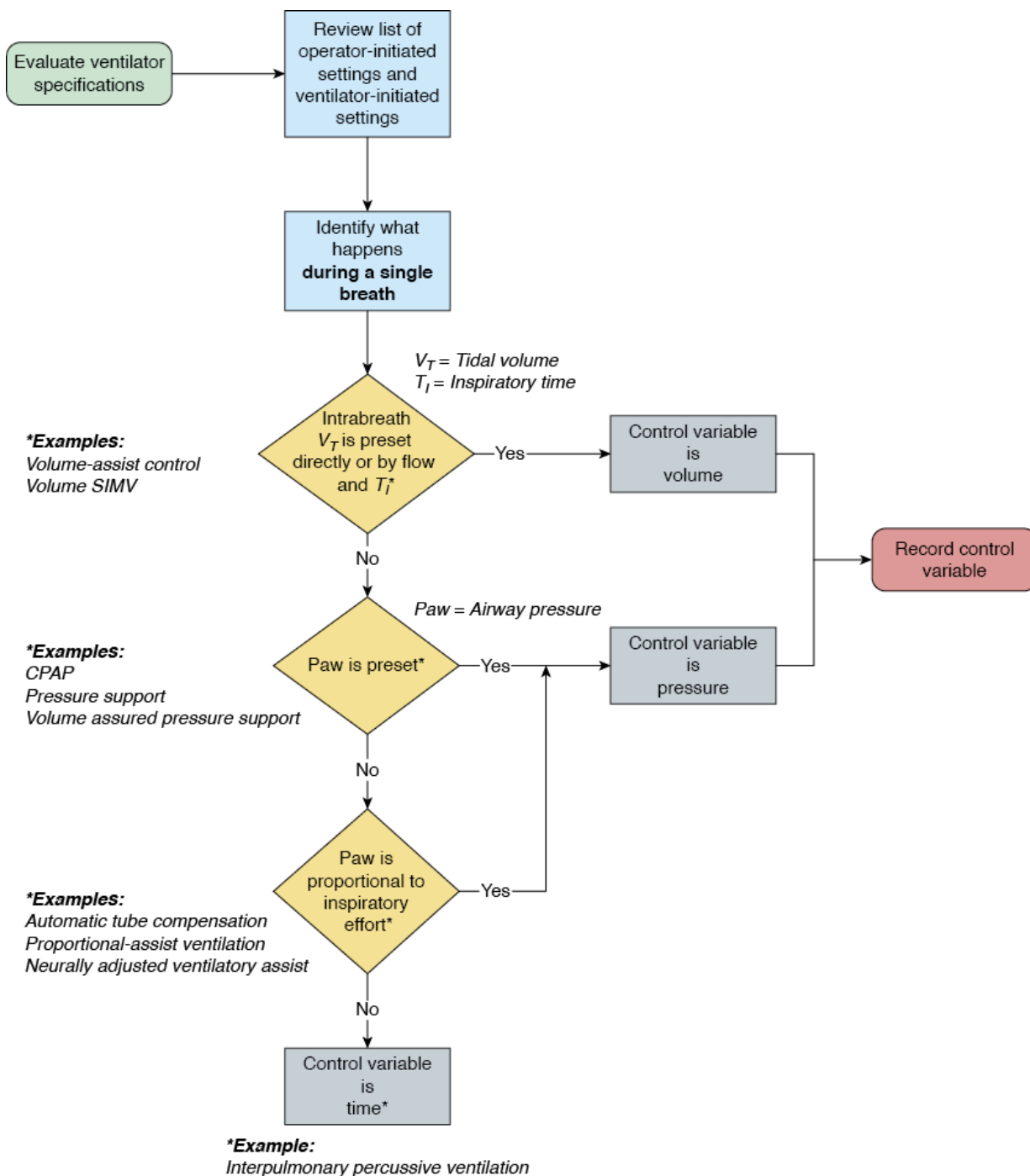
You can easily appreciate the motivation for classifying modes, just as we do animals or plants (or cars or drugs) because of their large number and variety. The logical basis for a mode taxonomy, however, is not apparent without some consideration. This basis has become a teaching system I have developed and tested and is founded on ten simple constructs (or aphorisms), each building on the previous one to yield a practical taxonomy. These aphorisms summarize many of the ideas discussed previously in this chapter, and there is even some evidence that they are recognized internationally by clinicians.³² In simplified form, the aphorisms are as follows:

1. A **breath** is one cycle of positive flow (inspiration) and negative flow (expiration). The purpose of a ventilator is to assist breathing. Therefore, the logical start of a taxonomy is to define a breath. Breaths are defined such that during mechanical ventilation, small artificial breaths may be superimposed on large natural breaths or vice versa.
2. A breath is **assisted** if pressure rises above baseline during inspiration or falls during expiration. A ventilator assists breathing by doing some portion of the work of breathing. This occurs by delivering volume under pressure.
3. A ventilator assists breathing using either **pressure control (PC)** or **volume control (VC)**. The equation of motion is the fundamental model for understanding patient–ventilator interaction and hence modes of ventilation. The equation is an expression of the idea that only one variable can be predetermined at a time; pressure or volume (flow control is ignored for simplicity and for historical reasons, and because controlling flow directly will indirectly control volume and vice versa).
4. Breaths are classified according to the criteria that **trigger** (start) and **cycle** (stop) inspiration. A ventilator must know when to start and stop flow delivery for a given breath. Because starting and stopping inspiratory flow are critical events in synchronizing patient–ventilator interaction, and because they involve uniquely different operator-influenced factors, they are distinguished by giving them different names.
5. Trigger and cycle criteria can be either patient or machine initiated. A major design consideration in creating modes is the ability to synchronize breath delivery with patient demand and at the same time to guarantee breath delivery if the patient is apneic. Therefore, understanding patient–ventilator interaction means understanding the difference between machine and patient trigger and cycle events.
6. Breaths are classified as **spontaneous** or **mandatory** based on both the trigger and cycle criteria. A spontaneous breath arises without apparent external cause. Thus, it is patient triggered and patient cycled. Any machine involvement in triggering or cycling leads to a mandatory breath. Note that the definition of a spontaneous breath is independent of the definition of an assisted or unassisted breath.
7. Ventilators deliver only three basic **breath sequences**: **CMV**, **IMV**, and **CSV**. The two breath classifications logically lead to three possible breath sequences that a mode can deliver. CSV implies all spontaneous breaths; IMV allows spontaneous breaths to occur between mandatory breaths and **CMV** does not.
8. There are only five basic **ventilatory patterns**: **VC-CMV**, **VC-IMV**, **PC-CMV**, **PC-IMV**, and **PC-CSV**. All modes can be categorized by these five patterns. This provides enough practical detail about a mode for most clinical purposes.
9. Within each ventilatory pattern there are several variations that can be distinguished by their **targeting scheme(s)**. When comparing modes or evaluating the capability of a ventilator, more detail is required than just the ventilatory pattern. Modes with the same pattern can be distinguished by describing the targeting schemes they use. There are at present only six basic targeting schemes: set-point, dual, servo, adaptive, optimal, and intelligent.
10. A **mode of ventilation** is classified according to its control variable, breath sequence, and targeting scheme(s). A practical taxonomy of ventilatory modes is based on just four levels of detail: the control variable (pressure or volume), the breath sequence (**CMV**, **IMV**, or **CSV**), the targeting scheme used for primary breaths (**CMV** and **CSV**), and, if applicable, secondary breaths (**IMV**).

In teaching these constructs to respiratory therapists and physicians, most educators would agree that knowing a concept and applying it are two different skills. As with any taxonomy, learning the definitions and mastering the heuristic thinking required to actually categorize specific cases requires further guidance and some practice. Say, for example, your task is to compare the capabilities of two major intensive care unit ventilator models for a large capital purchase. Memorizing the ten aphorisms may not translate into the ability to classify the modes offered on these two ventilators as a basis for comparison. To facilitate that skill, I created the three tools shown in [Figures 2-11 and 2-12](#) and in [Table 2-4](#). Using these tools you can create a simple spreadsheet that defines and compares the modes on any number of ventilators. [Table 2-5](#) is an example of such a table for the Covidien PB 840 ventilator and the Dräger Evita XL ventilator. When implemented as a spreadsheet with built-in data-sorting functions, the table becomes a database with several major uses:

1. A “Rosetta Stone” that can be used to translate from mode name to mode classification and vice versa. In this way modes can be identified that are functionally identical but have different proprietary names.
2. A tool for engineers to describe performance characteristics of individual named modes. Information like this should be available to users in the ventilator’s manual.
3. A system for clinicians to compare and contrast the capabilities of various modes and ventilators.
4. A paradigm for educators to use in teaching the basic principles of mechanical ventilation.

Figure 2-11

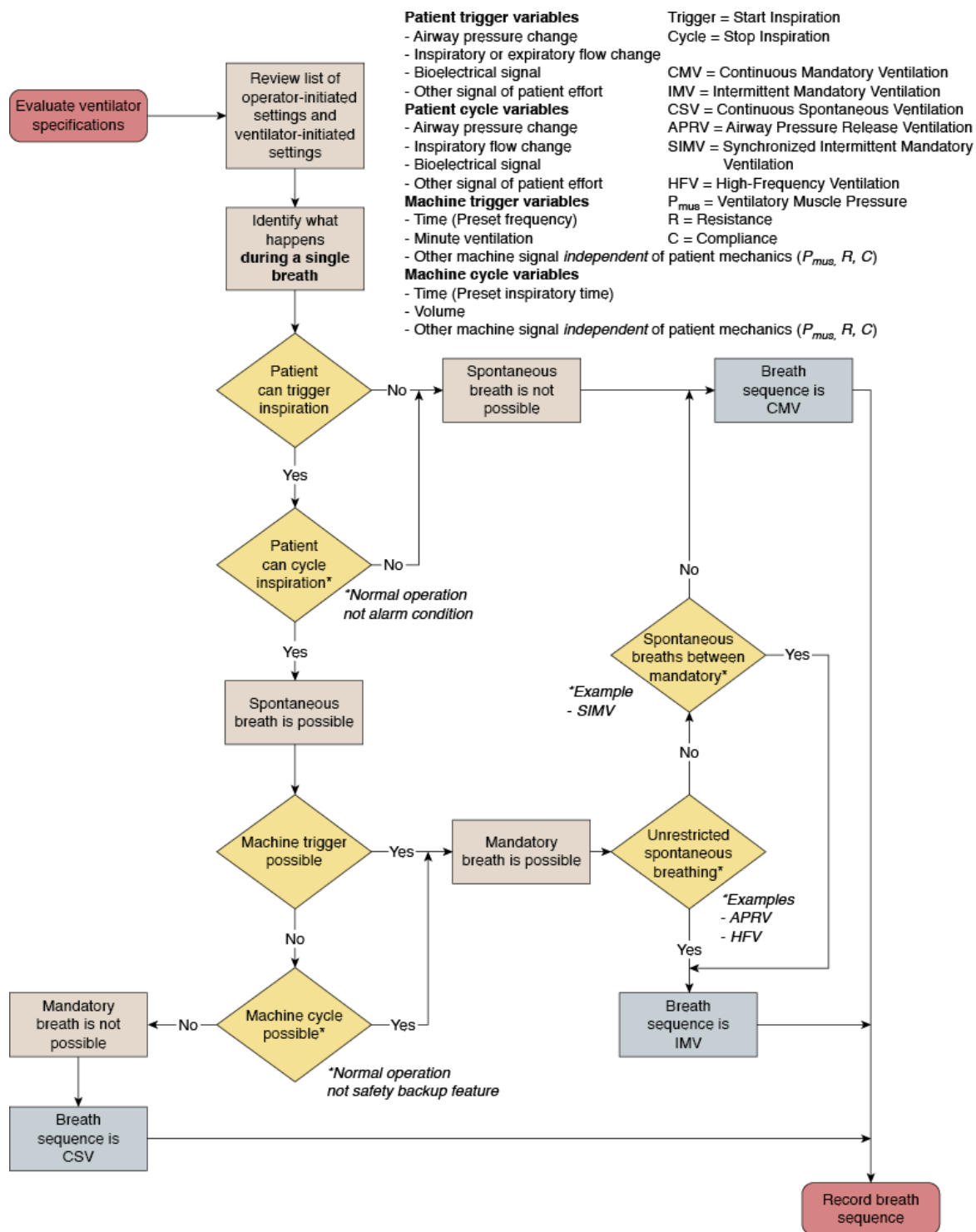


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Algorithm for determining the control variable when classifying a mode. SIMV, synchronized intermittent mandatory ventilation. (Copyright 2011 by Mandu Press Ltd. and reproduced with permission.)

Figure 2-12



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Algorithm for determining the breath sequence when classifying a mode. (Copyright 2011 by Mandu Press Ltd. and reproduced with permission.)

Table 2-4: Explanation of How Targeting Schemes Transform Operator Inputs into Ventilator Outputs

#	Control Variable	Target Scheme	Explanation	Example Mode Name	Predetermined Inputs			Ventilator Output	
					WB Target	Cycle	BB Target	+ Impedance	- Impedance
1	P	Set-point	Peak airway pressure is independent of impedance	PC SIMV	P	T			
2	P	Set-point	Peak airway pressure is independent of impedance	Pressure support	P	F			
3	P	Set-point	Peak airway pressure is independent of impedance	Automatic resuscitator	F	P			
4	V	Set-point	Tidal volume is independent of impedance	VC A/C	F	T			
5	P	Dual P-F	Same as #1 if secondary target not activated	VAPS	P,F	V			
6	V	Dual F-P	Same as #4 if secondary target not activated	CMV + Pressure Limited	F,P	V			
7	P	Servo	Pressure is automatically proportional to inspiratory effort Effort is represented by patient: flow volume and flow	ATC PAV+	Percent Support	F			
8	P	Servo	Pressure is automatically proportional to inspiratory effort represented by diaphragm EMG	NAVA	cm H ₂ O μv	NA			
9	P	Adaptive	Same as #1 within a breath plus volume target between breaths	PRVC	NA	T	Volume		
10	P	Optimal	Same as #9 plus algorithm to minimize inspiratory work rate	ASV	NA	F	%MV Frequency Volume		
11	P	Intelligent	Same as #9 plus volume, PCO ₂ and frequency targets using artificial intelligence algorithms	Smart Care/PS	NA	NA	Frequency Volume P _{ET} CO ₂		

P, pressure; V, volume; F, flow; T, time; R, resistance; E, elastance; MV, minute volume; Edi, electrical activity of diaphragm; WB Target, within-breath preset parameters of the pressure, volume, or flow waveform; BB Target, between breath targets modify WB targets or overall ventilatory pattern; Cycle, end of inspiration; NA, not available as operator preset, ventilator determines value if applicable.

low impedance, low resistance and/or elastance;

high impedance, high resistance and/or elastance;

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Table 2-5: Spreadsheet Example of How Modes on Two Common ICU Ventilators Would Be Classified

The spreadsheet could be sorted any number of ways (e.g., using AutoFilter drop-down dialogs) to compare the ventilators on various capabilities (e.g., all modes with adaptive pressure targeting). The spreadsheet also functions as a mode translator, giving the different proprietary names for identical modes.

			Order	Family	Genus	Species
					Primary Breath	Secondary Breath
Manufacturer	Model	Manufacturer's Mode Name	Primary Control Variable	Breath Sequence	Target Scheme	Target Scheme
Covidien	840	Volume Control Plus Assist Control	Pressure	CMV	adaptive	N/A
Covidien	840	Volume Support	Pressure	CSV	adaptive	N/A
Covidien	840	Volume Control Plus Synchronized Intermittent Mandatory Ventilation	Pressure	IMV	adaptive	set-point
Covidien	840	Volume Ventilation Plus Synchronized Intermittent Mandatory Ventilation	Pressure	IMV	adaptive	adaptive
Covidien	840	Tube Compensation	Pressure	CSV	servo	N/A
Covidien	840	Proportional Assist Plus	Pressure	CSV	servo	N/A
Covidien	840	Pressure Control Assist Control	Pressure	CMV	set-point	N/A
Covidien	840	Pressure Support	Pressure	CSV	set-point	N/A
Covidien	840	Spontaneous	Pressure	CSV	set-point	N/A
Covidien	840	Pressure Control Synchronized Intermittent Mandatory Ventilation	Pressure	IMV	set-point	set-point
Covidien	840	BiLevel	Pressure	IMV	set-point	set-point
Covidien	840	Volume Control/Assist Control	Volume	CMV	set-point	N/A
Covidien	840	Volume Control Synchronized Intermittent Mandatory Ventilation	Volume	IMV	set-point	set-point
Dräger	Evita XL	Mandatory Minute Volume with AutoFlow	Pressure	IMV	adaptive	set-point
Dräger	Evita XL	Continuous Mandatory Ventilation with AutoFlow	Pressure	CMV	adaptive	N/A
Dräger	Evita XL	Synchronized Intermittent Mandatory Ventilation with AutoFlow	Pressure	IMV	adaptive	set-point
Dräger	Evita XL	SmartCare	Pressure	CSV	intelligent	N/A
Dräger	Evita XL	Automatic Tube Compensation	Pressure	CSV	servo	N/A
Dräger	Evita	Pressure Controlled Ventilation Plus Assisted	Pressure	CMV	set-point	set-point

	XL					
Dräger	Evita XL	Pressure Controlled Ventilation Plus Pressure Support	Pressure	IMV	set-point	set-point
Dräger	Evita XL	Airway Pressure Release Ventilation	Pressure	IMV	set-point	set-point
Dräger	Evita XL	Continuous Positive Airway Pressure/Pressure Support	Pressure	CSV	set-point	N/A
Dräger	Evita XL	Mandatory Minute Volume	Volume	IMV	adaptive	set-point
Dräger	Evita XL	Continuous Mandatory Ventilation with Pressure Limited Ventilation	Volume	CMV	dual	N/A
Dräger	Evita XL	Synchronized Intermittent Mandatory Ventilation with Pressure Limited Ventilation	Volume	IMV	dual	set-point
Dräger	Evita XL	Mandatory Minute Volume with Pressure Limited Ventilation	Volume	IMV	dual/adaptive	set-point
Dräger	Evita XL	Continuous Mandatory Ventilation	Volume	CMV	set-point	N/A
Dräger	Evita XL	Synchronized Intermittent Mandatory Ventilation	Volume	IMV	set-point	set-point

CMV, continuous mandatory ventilation; CSV, continuous spontaneous ventilation; IMV, intermittent mandatory ventilation.

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One can imagine the utility of an expanded database containing the classification of all modes on all commercially available ventilators.

Ventilator Alarm Systems

As with other components of ventilation systems, ventilator alarms have increased in number and complexity. Fortunately, the classification system I have been describing can be expanded to include alarms as well (see [Table 2-1](#)).

MacIntyre³³ has suggested that alarms also be categorized by the events that they are designed to detect. Level 1 events include life-threatening situations, such as loss of input power or ventilator malfunction (e.g., excessive or no flow of gas to the patient). The alarms in this category should be mandatory (i.e., not subject to operator choice), redundant (i.e., multiple sensors and circuits), and noncanceling (i.e., alarm continues to be activated, even if the event is corrected, and must be reset manually). Level 2 events can lead to life-threatening situations if not corrected in a timely fashion. These events include such things as blender failure, high or low airway pressure, autotriggering, and partial patient circuit occlusion. They also may include suspicious ventilator settings such as an inspiratory-to-expiratory timing (I:E) ratio greater than 1:1. Alarms for level 2 events may not be redundant and may be self-canceling (i.e., alarm inactivated if event ceases to occur). Level 3 events are those that affect the patient-ventilator interface and may influence the level of support provided. Examples of such events are changes in patient compliance and resistance, changes in patient respiratory drive, and auto-PEEP. Alarm function at this level is similar to that of level 2 alarms. Level 4 events reflect the patient condition alone rather than ventilator function. As such, these events usually are detected by stand-alone monitors, such as oximeters, cardiac monitors, and blood-gas analyzers. Some ventilators, however, are able to incorporate the readings of a capnograph in their displays and alarm systems.

The Future

Almost 20 years ago, Warren Sanborn predicted that ventilators today would "... report the patient's metabolic state; manage oxygen delivery; calculate cardiac output, synchronize breath delivery with cardiac cycle to maximize cardiac output...and perform all these functions automatically or at least

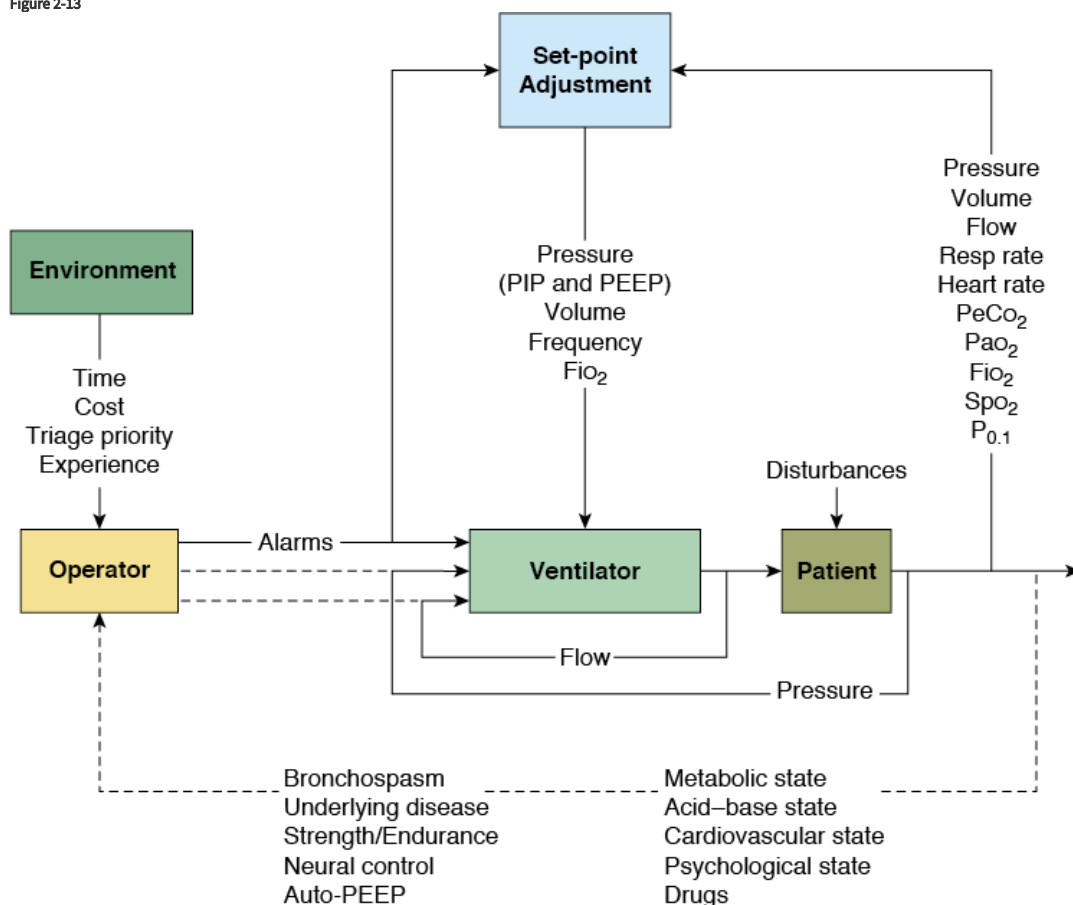
presenting consensus-based advisory messages to the practitioner....¹⁷ Some of these ideas were never developed commercially. Some were tried and abandoned. Some, have evolved beyond Warren's broad vision.

There are three basic ways to improve ventilators in the future. First, just like computer games, ventilators need to improve the operator interface constantly. Yet very little research has been done to call attention to problems with current displays.^{34,35} We have come a long way from using a crank to adjust the stroke of a ventilator's piston to set tidal volume. The operator interface must provide for three basic functions: allow input of control and alarm parameters, monitor the ventilator's status, and monitor the ventilator-patient interaction status. We have a long way to go before the user interface provides an ideal experience with these functions.

Second, the weak link in the patient-ventilator system is the patient circuit. We buy a \$35,000 ventilator with state-of-the-art computer control, and then we connect it to the patient (priceless) with a \$1.98 piece of plastic tubing that is subject to filling with condensate from a heated humidifier whose design has not changed appreciably in 20 years. The resistance and compliance of the delivery circuit make flow control and volume delivery more difficult. It is like buying a Ferrari and putting wooden wheels on it. In the future, water vapor should be treated like any other desirable inhaled gas constituent (e.g., air, oxygen, helium, or nitric oxide) and metered from within the ventilator. The inspiratory part of the patient circuit should be a sterile, insulated, permanent part of the ventilator right up to the patient connection, which can be a disposable tip for cleaning purposes. The gas should be delivered under high pressure as a jet to provide not only conventional pressure, volume, and flow waveforms but also high-frequency ventilation. The jet also can be used to provide a counterflow PEEP effect, eliminating any need for an exhalation-valve system. The disposable tip could be designed to house disposable sensors and would be the only part of the circuit to be exposed to the patient's exhaled gas. If ventilator manufacturers saw themselves as providers of the entire system, instead of letting third parties deal in plastic connecting tubing, I think we would see a huge evolutionary step in ventilator performance, better patient outcomes, and potential savings in labor costs for providers.

Third, the most exciting area for development probably is in the intelligence that will be built into future ventilator control circuits. The real challenge in closed-loop control of ventilation is defining, measuring, and interpreting the appropriate feedback signals. If we stop to consider all the variables a human operator assesses, the problem looks insurmountable. Not only does a human consider a wide range of individual physiologic variables, but there are the more abstract evaluations of such things as metabolic, cardiovascular, and psychological states. Add to this the various environmental factors that may affect operator judgment, and we get a truly complex control problem (Fig. 2-13).

Figure 2-13



The challenge of total computer control of mechanical ventilation. *Solid arrows* depict signals that have been used at least experimentally. *Dotted arrows* represent potential feedback signals. (Reproduced, with permission, from Chatburn RL. Computer control of mechanical ventilation. *Respir Care*. 2004;49:507–515.)

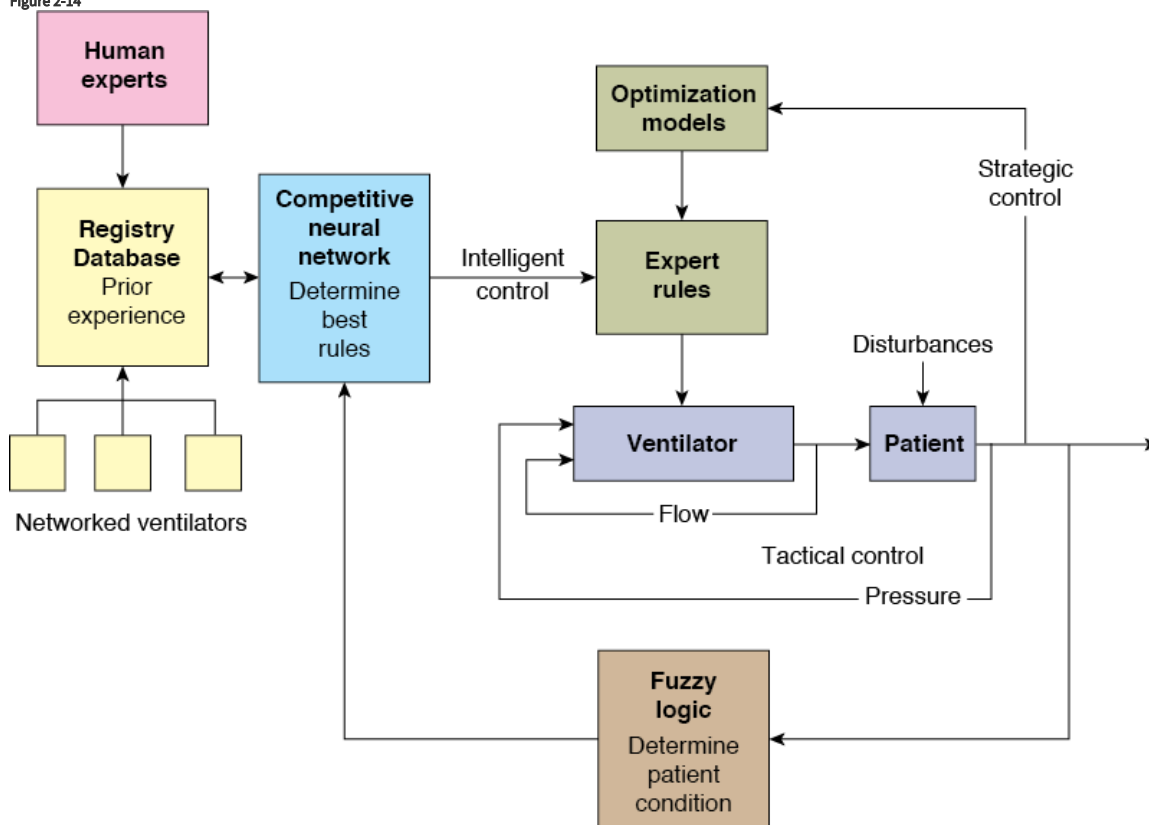
I would like to speculate now about a response to this challenge. The ideal control strategy would have to start out with basic tactical control of the individual breath. Next, we add longer-term strategic control that adapts to changing load characteristics. Mathematical models could provide the basic parameters of the mode, whereas expert rules would place limits to ensure lung protection.

Next, we sample various physiologic parameters and use fuzzy logic to establish the patient's immediate condition. This information is passed on to a neural network, which would then select the best response to the patient's condition.

The neural network ideally would have access to a huge database comprised of both human expert rules and actual patient responses to various ventilator strategies. This arrangement would allow the ventilator not only to learn from its interaction with the current patient but also to contribute to the database.

Finally, the database and this ventilator could be networked with other intelligent ventilators to multiply the learning capacity exponentially (Fig. 2-14). Whatever the future brings, it seems clear that ventilators will have more intelligence built in to increase patient safety and decrease the time required to provide care.

Figure 2-14



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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A potential approach to the challenge of fully automated control of mechanical ventilation. (Reproduced, with permission, from Chatburn RL. Computer control of mechanical ventilation. *Respir Care*. 2004;49:507–515.)

Summary and Conclusion

Mechanical ventilators have become so complex that a system of classification is necessary to communicate intelligently about them. The theoretical basis for this classification system is a mathematical model of patient–ventilator interaction known as the *equation of motion for the respiratory system*. From this model we deduce that as far as an individual inspiration is concerned, any conceivable ventilator can be classified as either a pressure, volume, or flow controller (and in rare cases, simply an inspiratory-expiratory time controller). An individual breath is shaped by the phase variables that determine how the breath is triggered (started), targeted (sustained), and cycled (stopped).

A *mode* of ventilation can be characterized using a four-level taxonomy: (a) control variable, that is, pressure or volume according to the equation of motion; (b) the breath sequence, that is, **CMV**, IMV, or CSV; (c) targeting scheme for primary breaths; and (d) targeting scheme for secondary breaths. The

trend in ventilator targeting schemes has been from basic manual control (within-breath control requiring operator input of static set-points), to more advanced automatic control (between-breath control of set-points that are adjusted automatically by the ventilator with minimal operator input), to the highest level of intelligent control (in which the operator theoretically may be eliminated altogether in favor of artificial-intelligence systems capable of learning).

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